The Double-Ended Helix and Projection Primacy: A Machine-Checked Spectral Realization for Dirichlet L-Functions, with the Generalized Riemann Hypothesis as the Completeness of its Projection

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2026-07-01 (major revision; v1 2026-06-19)

Abstract

We present a geometric spectral realization for Dirichlet L-functions — a three-dimensional, double-ended *Frobenius-dual helix carrier* whose phasor fiber accumulates L — together with a machine-checked equivalence isolating exactly where the Generalized Riemann Hypothesis enters. Three theorems carry the paper. *Unconditional realization*: every zero of $L(s, \chi)$ on the critical line is realized as an admissible real source-height eigen-event $Z = e^{\gamma} > 0$ of a Hermitian positive-semidefinite pencil family — the unsigned channel alive, the signed channel closed, the Gram determinant rank-dropping by exactly one (`nontrivial_zero_represented`, `no_double_cancellation`). *Unconditional converse implication*: any nontrivial cancellation realized as a kernel of the fixed self-adjoint generator lies on the critical line, by von Neumann reality (`GRH_from_nontrivial_Aχ_realization`). *The isolated gap*: whether *all* nontrivial zeros are so realized — the *projection-primacy principle* — is proved equivalent to the real-height exhaustion of the zeros, hence to GRH for χ (`projectionPrimacy_iff_exhausts`); it is neither assumed nor proved. The Riemann Hypothesis is thus reframed, not approached: it is exactly the completeness of the carrier's projection.

The geometry beneath these statements is derived, not designed. Uniform $\pi/3$-arclength placement on the growing-radius helix forces the area law $r_n \sim C\sqrt{n}$ (`windIntegerSite_radius_sq_tendsto`), and $\sigma = \frac{1}{2}$ is the unique scale-balanced exponent (`sigma_half_is_scale_critical`): the abscissa of the critical line comes from the carrier's scale geometry before any zero-location statement enters. The analytic fiber is exact: for non-principal χ the partial sums $\sum_{n<N} \chi(n)n^{-s}$ converge to $L(s, \chi)$ on all of $\mathrm{Re}\, s > 0$ (`dirichlet_strip_tendsto_LFunction`), and the η-regularized principal channel identifies ζ's on-line vanishings exactly (`FullFiber_eq_zero_iff_zeta`) — identities with no cutoff error, machine-checked.

The model is also an experimental instrument, and we report its phenomenology. The partial-absorption law (each phasor completes exactly at its own integer site, one at a time — the finite shadow of `no_double_cancellation`) reproduces the focal cancellation at *all 111 critical-line zeros below height 36* of the eight real primitive Dirichlet characters of conductor ≤ 12, at eigenheights up to $N = e^{\gamma} \approx 3.8 \times 10^{15}$, with residuals at the truncation scale $\sim N^{-1/2}$; an error budget separates zero-precision, truncation, and evaluator floors. The same machinery, with lanes given by coefficient signs instead of residue classes, locates the zeros of degree-two L-functions (Δ and the level-11 elliptic newform) — the mechanism does not depend on character periodicity. Negative results are reported alongside.

Every named statement is formalized in Lean 4 over Mathlib and checked by the kernel: no `sorry`, no construction-specific axiom; the axiom footprint of the named theorems is {propext, Classical.choice, Quot.sound}. Appendix A maps every headline claim to its Lean identifier.

Contents

1 Main results

The paper has two halves: a body of theorems (Sections 3–15), every one machine-checked, and a body of computational phenomenology (Section 16) validating the geometric mechanism at scale. Five results carry the mathematical half; they are easily blurred if the helix picture is read only as a visualization.

(1) *Scale geometry (Section 3).* The $\sqrt{n}$ radius is not stipulated. Integers are placed at fixed $\pi/3$ arclength spacing and wound onto the helix; the resulting radius satisfies $r_n^2/n \to r(\pi/3)/\pi$ (Theorem 3.9), and $\sigma = \frac{1}{2}$ is the *unique* scale-balanced exponent (Theorem 3.11). The abscissa of the critical line is derived from the carrier geometry before any zero-location statement enters.

(2) *Exact analytic accumulation (Section 7).* For non-principal characters the phasor channel converges exactly to $L(s,\chi)$ on all of $\operatorname{Re} s > 0$ (Theorem 7.1); the η-regularized principal channel identifies ζ's on-line vanishings exactly (Theorems 7.2, 7.3, 7.5). The fiber has an exact value in the strip — no finite-window approximation, no remainder term.

(3) *Unconditional realization (Sections 13, 14).* Every critical-line zero of $L(\cdot,\chi)$ is realized as an admissible real source-height eigen-event $Z = e^{\gamma} > 0$: unsigned channel alive, signed channel closed, and the Hermitian positive-semidefinite harmonic Gram pencil rank-drops — by exactly one, never to the zero matrix, and calibration-independently across the whole pencil family (Theorem 13.5; `nontrivial_zero_represented`, `gram_rank_drop_calibration_independent`, `no_double_cancellation`).

(4) *Self-adjointness forces the line (Section 14).* Conversely, any nontrivial cancellation realized as a kernel event of the fixed self-adjoint generator A_χ has $\operatorname{Re} s = \frac{1}{2}$, by von Neumann reality (Theorem 14.2). Together with (3) this is a two-sided, unconditional Hilbert–Pólya-type package for the critical-line zeros.

(5) *The isolated gap is exactly GRH (Sections 14, 15).* Whether *all* nontrivial zeros are so realized — *projection primacy* — is proved equivalent to the real-height exhaustion of the zeros, hence to GRH for χ (Theorem 14.3, Remark 15.7). It is stated as an explicit named hypothesis, neither assumed nor proved; conditional on it, every nontrivial zero is a represented eigen-event.

The phenomenology half reports the model as an instrument: the partial-absorption law and its validation at all 111 critical-line zeros below height 36 of the eight real primitive characters of conductor ≤ 12 (eigenheights to 3.8×10^{15}), a measured error budget, the degree-two (Δ, level-11 newform) extension showing the mechanism is not an artifact of character periodicity, and the negative results. Section 17 collects structural frameworks (chiral cup, Kähler polarization, de Branges evaluation) as machine-checked scaffolding for future work, deliberately demoted from the main line. Section 18 records the verification; Appendix A maps the headline claims to Lean identifiers.

2 Objects and notation

Let χ be a Dirichlet character modulo q and $L(s, \chi)$ the associated L-function; ζ is the Riemann zeta function and $\Lambda(s) = \texttt{completedRiemannZeta}(s)$ its completed form, normalized so that $\Lambda(1-s) = \Lambda(s)$. The *critical line* is the set $s = \frac{1}{2} + iy$, $y \in \mathbb{R}$. We write $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ for the unit circle (Mathlib `Circle`) and $\overline{\cdot}$ for complex conjugation (`starRingEnd` $\mathbb{C}$). For an arithmetic weight χ the *phasor* attached to the integer $n \geq 1$ on the vertical line $s = \sigma + iy$ is $\chi(n)\, n^{-s}$.

The abscissa $\frac{1}{2}$*.* Every $\frac{1}{2}$ below is one of two *derived* constants, neither of which assumes where the zeros lie. As the carrier's *scale-critical exponent* it is the unique σ at which the phasor amplitude $n^{-\sigma}$ balances the emergent area-law radius $r_n \sim \sqrt{n}$ (Theorem 3.11); this fixes the amplitude $n^{-1/2}$ and the abscissa of the line the fiber rides, from the carrier geometry alone. As the *reflection axis* $\operatorname{Re} s = \frac{1}{2}$ it is the fixed line of $s \mapsto 1 - s$, on which $\overline{s} = 1 - s$; with the functional equation $\Lambda(1-s) = \Lambda(s)$ this is what makes the completed object real on the line (Theorem 9.3) and pairs the two chiralities (Theorem 11.3). The Riemann Hypothesis — that the *zeros* ρ satisfy $\operatorname{Re} \rho = \frac{1}{2}$ — is a third, distinct statement, never assumed here: the on-line results below are either unconditional in y or conditional on a zero being present at the stated height, and they assert nothing about off-line zeros.

The classical analytic theory of ζ and Dirichlet L-functions — the Euler product, the functional equation, and the von Mangoldt explicit formula [1, 2, 3, 4, 5, 7] — enters here through the corresponding Mathlib lemmas.

All named results carry the identifier of the corresponding Lean declaration, e.g. `frobenius_conjugate_det_one`; Section 18 records the verification.

3 The $\pi/3$ helix carrier

3.1 The space curve and its arclength

Definition 3.1 (Carrier curve; `Geometry.helix`, `helixVel`, `speed`)**.** *For pitch p and radial rate r, the carrier is the growing-radius helix*

$$\gamma(k) \;=\; \big(r\,k\cos 2\pi k,\; r\,k\sin 2\pi k,\; p\,k\big), \qquad k \in \mathbb{R}.$$

Its velocity is $\gamma'(k) = \big(r\cos 2\pi k - 2\pi rk\sin 2\pi k,\ r\sin 2\pi k + 2\pi rk\cos 2\pi k,\ p\big)$*, and the squared speed is*

$$\|\gamma'(k)\|^2 \;=\; p^2 + r^2 + (2\pi rk)^2.$$

The climber *is* $k(y) = e^y/p$*, so the height is* $z = p\,k(y) = e^y$ *and the cylindrical radius is* $R(y) = r\,k(y) = (r/p)e^y$*.*

Remark 3.2 (Constant height law: $\Delta z = 1$ per integer n, not per phasor; `Geometry.heightLaw`, `heightLaw_step`, `resonance_height`, `activeFraction_eq_totient`, `carrierRadius_growth_div_sqrt_tendsto`)**.** *Discretely, the carrier assigns integer* n *the height* $z_n = n$*: a constant* $\Delta z = 1$ per integer n *(`heightLaw_step`:* heightLaw$(n+1) -$ heightLaw$(n) = 1$*)—every slot, whether or not its arithmetic weight* $\chi(n)$ *is nonzero. A neutral slot (*$\chi(n) = 0$*) carries no arrow yet still occupies its place and still advances the height. Counting instead per* active *phasor would climb by the active fraction* $\varphi(q)/q$ *per integer (`activeFraction_eq_totient`: only the* $\varphi(q)$ *unit residues carry a spinning arrow) and land the resonance at* $(\varphi(q)/q)\,e^{\gamma}$*, short by exactly that fraction. Since phasor* n *resonates at* $y = \log n$*, integer* n *sits at* $z_n = n = e^{\log n}$*—the height of its own resonant frequency (`resonance_height`)—so a zero at frequency* γ *(resonant term* $n^* = e^{\gamma}$*) lands at height* $z = e^{\gamma}$*, for every L-function (it rides only on* spin $= \log n$*; the located on-line vanishings are exactly the carrier's resonant heights, `online_zero_eq_carrierZeros`, each at a unique height, `online_zero_unique_height`). This is the discrete form of the climber height* $z = p\,k(y) = e^y$ *above. The radial gap per turn is L-dependent,* $g = e^{(q \bmod)}$ *(eta* e^2*,* χ_3 e^3*, ...), so the area-law radius specializes to* $r_n/\sqrt{n} \to \sqrt{e^q/3}$ *(`carrierRadius_growth_div_sqrt_tendsto`): the same* $\sqrt{n}$ *shape for every L, only the constant differs — consistent with the gauge-independence of the scale-critical exponent (Theorem 3.11).*

Definition 3.3 (Arclength; `Geometry.arclength`, `arclengthClosed`)**.** *The arclength of* γ *from* 0 *to* k *is* $S(k;p,r) = \displaystyle\int_0^k \sqrt{p^2 + r^2 + (2\pi rt)^2}\,dt$*.*

Theorem 3.4 (Closed-form arclength; `arclength_closed_form`, `arclength_r_zero`)**.** *For* $r > 0$*,*

$$S(k;p,r) \;=\; \frac{k}{2}\sqrt{p^2 + r^2 + 4\pi^2 r^2 k^2} \;+\; \frac{p^2 + r^2}{4\pi r}\ \operatorname{arsinh}\!\Big(\frac{2\pi rk}{\sqrt{p^2 + r^2}}\Big),$$

and $S(k;p,0) = p\,k$ *for* $p \ge 0$*.*

Proof sketch. Differentiate the stated antiderivative and check it equals the integrand $\sqrt{p^2 + r^2 + 4\pi^2 r^2 k^2}$ pointwise, then apply the fundamental theorem of calculus on $[0,k]$; the constant-pitch case is $\int_0^k p\,dt$. ☐

3.2 Uniform integer placement and the area law

Definition 3.5 (Spacing, index, integer angle; `Geometry.Delta`, `Nindex`, `spinAngle`)**.** *The fixed spacing is* $\Delta = \pi/3$*. The continuous geometric index is* $N(y) = S(k(y);p,r)/\Delta$*, so* $N(y) = (3/\pi)\,S(k(y);p,r)$ *(`Nindex_eq`). The integer angular coordinate is* $s_n = n \cdot (\pi/3)$*.*

Theorem 3.6 (Uniform $\pi/3$ spacing; `carrier_spacing`)**.** *For every* $n \in \mathbb{N}$*,* $s_{n+1} - s_n = \pi/3$*.*

Theorem 3.7 (Area-law bound; `arclengthClosed_lower_bound`, `arclengthClosed_upper_bound`)**.** *For* $r > 0$ *and* $k \ge 0$*,*

$$\pi r\,k^2 \;\le\; S(k;p,r) \;\le\; \pi r\,k^2 + 2k\sqrt{p^2 + r^2}.$$

Definition 3.8 (Wound integer placement; `Geometry.windParameter`, `windIntegerSite`, `exists_unique_windParameter`). *Integer n sits at arclength $s_n = n\Delta$ on the unwound line (Definition 3.5).* Winding the line onto the helix *places it at the unique parameter $k_n \geq 0$ with $S(k_n; p, r) = n\Delta$ (`exists_unique_windParameter`; the map $s \mapsto k_n$ is `windParameter`). The wound site is* windIntegerSite$(n) = \gamma(k_n)$*, with cylindrical radius $r_n = r\, k_n$ (`windIntegerSite_cyl_radius`).*

Theorem 3.9 (Emergent radius / area law; `emergent_radius_sq_tendsto`, `windIntegerSite_radius_sq_tendsto`
For the wound placement $S(k_n; p, r) = n\Delta$ of Definition 3.8,

$$\frac{r_n^2}{n} = \frac{(r\, k_n)^2}{n} \xrightarrow[n\to\infty]{} \frac{r\Delta}{\pi}, \qquad \textit{equivalently} \qquad r_n \sim \sqrt{\tfrac{r\Delta}{\pi}}\,\sqrt{n},$$

so the enclosed area $\pi r_n^2 \sim r\Delta\, n$ is linear in n. The factor $\sqrt{n}$ is thus derived from the arclength geometry*: the bound of Theorem 3.7 forces $S(k) \propto k^2$, hence $k_n \propto \sqrt{n}$ and $r_n = r\, k_n \propto \sqrt{n}$ — it is not posited.*

Remark 3.10 (Unit gauge; `windIntegerSite_radius_sq_tendsto_unit_gauge`). *The area-law constant is $r\Delta/\pi$; with the native spacing $\Delta = \pi/3$ it equals $r/3$, so $r_n \sim \sqrt{n}$* exactly *in the gauge $r\Delta = \pi$, i.e. $r = 3$ (`windIntegerSite_radius_sq_tendsto_unit_gauge`); in the alternative gauge $r = 1$ the constant is $1/\sqrt{3}$. In this unit gauge the carrier point* helixPt$(n) = \sqrt{n} \cdot$ wind(n) *(Definition 4.1) has modulus* $\|$helixPt$(n)\| = \sqrt{n}$ *(`HelixLogFree.norm_helixPt`), matching the emergent radius r_n asymptotically (`helixPt_radius_matches_areaLaw`: the ratio* $\|$helixPt$(n)\|/r_n \to 1$ *in this gauge); the winding contributes unit modulus (Theorem 4.2).*

3.3 The scale-critical exponent

The area law gives more than the radius $\sqrt{n}$: it pins the critical exponent. Write $r_n =$ carrierRadius(n) for the wound radius of Definition 3.8 (`Geometry.carrierRadius`), and pair it with a candidate weight $n^{-\sigma}$, $\sigma \in \mathbb{R}$.

Theorem 3.11 (The critical exponent is scale-critical; `sigma_half_is_scale_critical`). *For $\sigma \in \mathbb{R}$, the scale-balance product $n^{-\sigma} r_n$ has a positive limit if and only if $\sigma = \frac{1}{2}$:*

$$\big(\exists L > 0:\ n^{-\sigma}\, r_n \xrightarrow[n\to\infty]{} L\big) \qquad \Longleftrightarrow \qquad \sigma = \tfrac{1}{2}.$$

Concretely, $n^{-\sigma} r_n \to 0$ for $\sigma > \frac{1}{2}$ (the weight decays faster than the radius grows), $n^{-\sigma} r_n \to +\infty$ for $\sigma < \frac{1}{2}$ (the weight outruns the radius), and $n^{-\sigma} r_n \to \sqrt{r\Delta/\pi} > 0$ for $\sigma = \frac{1}{2}$. Thus $\frac{1}{2}$ is the unique exponent at which the weight reciprocates the emergent radius. It is gauge-independent*: the gauge $r\Delta$ sets only the value of the limit, never which σ is critical.*

Proof sketch. By Theorem 3.9, $r_n/\sqrt{n} \to \sqrt{r\Delta/\pi} =: c > 0$. Factor $n^{-\sigma} r_n = (r_n/\sqrt{n}) \cdot n^{1/2-\sigma}$; the first factor tends to c, and the second tends to 0, to 1, or to $+\infty$ according as $\sigma > \frac{1}{2}$, $\sigma = \frac{1}{2}$, or $\sigma < \frac{1}{2}$. Only $\sigma = \frac{1}{2}$ yields a positive finite limit; uniqueness of limits closes both directions. □

Remark 3.12. *This is the geometric origin of the critical line: the $\frac{1}{2}$ of* $\operatorname{Re} s = \frac{1}{2}$ *is the $\frac{1}{2}$ of $r_n \sim \sqrt{n} = n^{1/2}$. It locates the line on which the phasor weights are scale-normalized to $O(1)$, so that the accumulation is governed by the winding (the phase cancellation) rather than the amplitude — which is exactly the mechanism by which the vanishing points arise (Section 7). It is a statement about the carrier alone; that the zeros of ζ and L then* lie *on this line is the separate analytic content of the functional equation (Theorem 11.3), not of the area law.*

3.4 The mod-6 bucket structure

Because $6 \cdot (\pi/3) = 2\pi$, the integer angle s_n is 6-periodic on the circle.

Theorem 3.13 (Bucket periodicity and counts; `spin_phasor_period6`, `bucket_count`)**.** $\exp(i\, s_{n+6}) = \exp(i\, s_n)$ *for all* n*, and for* $a < 6$ *exactly* N *of the integers in* $[0, 6N)$ *satisfy* $n \equiv a \pmod 6$*. Thus the integers collapse onto six angular directions, each receiving an equal share.*

Remark 3.14 (Sixth roots of unity; Eisenstein integers)**.** *The integer-angle phasor* $\exp(i\, s_n) = \exp(i\, n\pi/3)$ *takes values in the sixth roots of unity* $\mu_6 = \{e^{ik\pi/3} : 0 \le k < 6\}$*, cycling through them as* n *runs mod* 6*; the primitive sixth root* $\zeta = e^{i\pi/3}$ *(*$\zeta^2 = \zeta - 1$*, the cyclotomic relation* Φ_6*) generates* μ_6*. The ring* $\mathbb{Z}[\zeta]$ *is the ring of Eisenstein integers [8] — a hexagonal (triangular) lattice whose unit group is exactly* μ_6*. So the* $\pi/3$ *gauge is the Eisenstein symmetry, and the six buckets of Theorem 3.13 are its units.*

Theorem 3.15 (Exact finite monodromy of the $\pi/3$ gauge; `pi3_finite_monodromy`, `unit_not_finite_monodromy`, `circleMonodromy_iff_rat`)**.** *Read the carrier phase after* k *cells modulo a turn,* $C_H(k) = k\,H \pmod{2\pi}$*; a cell unit* H *has* exact finite circle monodromy *when the carrier returns to its origin after some positive number of cells, i.e.* $k\,H = 2\pi m$ *for integers* $k \ge 1$*,* $m \ge 1$*. The native gauge* $H = \pi/3$ *closes* exactly *after six cells,* $6 \cdot (\pi/3) = 2\pi$ *(*$k = 6$*,* $m = 1$*;* `pi3_finite_monodromy`*), whereas the unit gauge* $H = 1$ *never closes (*`unit_not_finite_monodromy`*: it would force* $\pi \in \mathbb{Q}$*). In general* H *has exact finite monodromy* iff $H/(2\pi) \in \mathbb{Q}_{>0}$ *(*`circleMonodromy_iff_rat`*). Thus* $\pi/3$ *is distinguished not only as the Eisenstein unit angle (Remark 3.14) but as a* finite-monodromy *gauge — the exact closing condition underlying the six buckets of Theorem 3.13.*

Remark 3.16 (A closed carrier for an open winding)**.** *The exactness of this closure is a genuine structural distinction from the object the carrier serves. The analytic phasor winding* $n^{-it} = e^{-it\log n}$ *rides incommensurate frequencies — the* $\log p$ *are* $\mathbb{Q}$*-linearly independent — so it never returns: the standard presentation of* L *has no closed angular structure. The* $\pi/3$ *carrier, by contrast, closes* exactly *after six cells, and only the rational-angle gauges close at all (Theorem 3.15). The carrier thus supplies a closed, finite, commensurate angular home — the Eisenstein/*μ_6 *hexagonal lattice (Remark 3.14) — for an arithmetic that has none in its analytic form. (The cell enters the readout as a unit-modulus factor, divided out to recover the* L*-value in Section 7.)*

4 The log-free winding and the bridge

4.1 The fundamental-theorem-of-arithmetic winding

Definition 4.1 (Winding angle and winding; `HelixLogFree.windAngle`, `wind`, `helixPt`)**.** *Fix a prime-angle assignment* θ *(in Lean, a function* $\theta : \mathbb{N} \to \mathbb{R}$*; only its values on primes enter). The* winding angle *is the completely additive extension read off the prime factorization,*

$$\Theta(n) \;=\; \sum_{p^e \,\|\, n} e\,\theta(p) \;=\; \textit{n.factorization.sum}\,(\lambda\, p\, e.\, e \cdot \theta(p)),$$

and the winding *is* $\mathrm{wind}(n) = \exp(i\,\Theta(n)) \in \mathbb{T}$*. The carrier point is* $\mathrm{helixPt}(n) = \sqrt{n} \cdot \mathrm{wind}(n)$*. No logarithm occurs in* Θ*.*

Theorem 4.2 (Multiplicative character; `windAngle_mul`, `wind_mul`, `wind_one`)**.** *For* $m, n \neq 0$*,* $\Theta(mn) = \Theta(m) + \Theta(n)$ *and* $\mathrm{wind}(mn) = \mathrm{wind}(m)\,\mathrm{wind}(n)$*, with* $\mathrm{wind}(1) = 1$*. Thus* $\mathrm{wind} : \mathbb{N} \to \mathbb{T}$ *is a completely multiplicative character [6].*

Proof sketch. Θ is a sum over `Nat.factorization`; additivity is `Nat.factorization_mul` (the fundamental theorem of arithmetic). Exponentiating turns additivity into multiplicativity on $\mathbb{T}$ via `Circle.exp_add`. $\square$

4.2 The bridge $\mathrm{wind}\, n \leftrightarrow n^{it}$

The single place a logarithm enters is the external identification of the geometric winding with the analytic phasor.

Theorem 4.3 (Logarithm is FTA-additive; `real_log_eq_factorization_sum`). *For $n \neq 0$, $\log n = \sum_{p^e \,\|\, n} e \log p$.*

Theorem 4.4 (The bridge; `windAngle_glog`, `wind_glog_eq_cpow`). *With the prime-angle assignment $\theta(p) = \gamma \log p$, one has $\Theta(n) = \gamma \log n$ and hence*

$$\mathrm{wind}(n) \;=\; n^{i\gamma} \qquad (n \geq 1).$$

Proof sketch. Substitute Theorem 4.3 into Θ to get $\Theta(n) = \gamma \log n$; then $\exp(i\gamma \log n) = n^{i\gamma}$ by the principal-branch definition of $n^{i\gamma}$ for the positive real base n. $\square$

5 The phasor fiber and its accumulation

5.1 Phasor decomposition

Theorem 5.1 (Vertical-line phasor and magnitude; `cpow_vertical_line_phasor`, `norm_cpow_vertical_line`). *For a positive real base x and $s = \sigma + iy$,*

$$x^{-s} \;=\; x^{-\sigma} \exp\!\big(-(y \log x)\, i\big), \qquad \big|x^{-s}\big| \;=\; x^{-\sigma}.$$

In particular, on the critical line $n^{-(1/2+iy)} = n^{-1/2} \exp(-(y \log n) i)$ with magnitude $n^{-1/2}$ for every y (`cpow_critical_line`, `norm_cpow_critical_line`).

Definition 5.2 (The spin; `LFunctionPhasor.mellinSpin`). *The* spin *of the integer n at ordinate y is the unit-modulus factor* $\mathrm{spin}(y, n) = \exp(-(y \log n)\, i)$.

Theorem 5.3 (The fiber rides the carrier; `fiber_rides_carrier`). *Write* $\mathrm{wind}_{-t\log}$ *for the winding of Definition 4.1 with the assignment $\theta(p) = -t \log p$, so* $\mathrm{wind}_{-t\log}(n) = n^{-it}$ *by the bridge (Theorem 4.4). For $n \geq 1$, the carrier magnitude $n^{-1/2}$ — the reciprocal of the emergent radius $\sqrt{n}$ — times this winding is the Dirichlet phasor:*

$$n^{-1/2} \cdot \mathrm{wind}_{-t\log}(n) \;=\; n^{-(1/2+it)}.$$

Remark 5.4 (The fiber's exponent is scale-critical; `critical_exponent_is_scale_critical`). *The exponent $\frac{1}{2}$ here is not a chosen constant. By Theorem 3.11 — re-exported at the unit-gauge carrier $r = 3$ as `critical_exponent_is_scale_critical` — it is the unique σ for which the fiber amplitude $n^{-\sigma}$ scale-balances the area-law radius r_n. So the critical line $\mathrm{Re}\, s = \frac{1}{2}$ that the fiber rides is the scale-balance locus of the carrier, forced by the geometry rather than inserted.*

5.2 Accumulation to L

Definition 5.5 (Finite phasor carrier; `DirichletPhasorCarrier.finiteCarrier`, `phasorTerm`). *$G_{\chi,N}(s) = \sum_{n<N} \chi(n)\, n^{-s}$, the partial sums of the carrier-riding phasors.*

Theorem 5.6 (Accumulation on $\mathrm{Re}\, s > 1$; `fiber_accumulates_to_L`, `finiteCarrier_tendsto_LFunction`). *For every Dirichlet character χ and $\mathrm{Re}\, s > 1$, $G_{\chi,N}(s) \to L(s, \chi)$ as $N \to \infty$.*

6 The three-channel three-dimensional phasor

The planar phasor $\chi(n)\, n^{-\sigma} \exp(-(y \log n)i)$ of Section 5 is a 2D arrow whose length is $\|\chi(n)\|\, n^{-\sigma}$; on the *neutral* channel $\chi(n) = 0$ it collapses to 0 and the magnitude $n^{-\sigma}$ is lost. The carrier, by contrast, places *every* slot n (Remark 3.2), neutral or not. The reconciliation lifts each term to a genuine vector in $\mathbb{R}^3 = (\text{spin plane } \mathbb{R}^2) \times (\text{mass axis } \mathbb{R})$, modeled as $\mathbb{C} \times \mathbb{R}$.

Definition 6.1 (3D phasor and its three channels; `Phasor3D.phasor3D`, `channel`)**.**

$$\mathrm{phasor3D}(\chi, \sigma, y, n) = \Big(\underbrace{\chi(n)\, n^{-\sigma} \exp(-(y \log n)i)}_{\textit{spin plane}},\ \underbrace{(1 - \|\chi(n)\|)\, n^{-\sigma}}_{\textit{mass axis}} \Big).$$

The charged channels $\chi(n) = \pm 1$ place the rotating arrow $\pm n^{-\sigma} \exp(-(y \log n)i)$ in the spin plane with zero mass; the neutral channel $\chi(n) = 0$ places the real amplitude $n^{-\sigma}$ on the mass axis with no spin.

Theorem 6.2 (The spin plane carries L unchanged; `phasor3D_plane_tsum`)**.** *The spin-plane (complex) coordinate of* phasor3D *is exactly the planar phasor of Section 5, so its accumulation is the same Dirichlet L-series. The 3D lift changes nothing in the analytic representation — it only records what the plane discards.*

Theorem 6.3 (Magnitude conservation across all channels; `phasor3D_mag3`)**.** *For a quadratic character the full $\mathbb{R}^3$ Euclidean length of every term is exactly $n^{-\sigma}$, on* all three *channels including the neutral one — where the planar length $\|\chi(n)\|\, n^{-\sigma}$ vanished. The magnitude lost from the plane is precisely the mass stored on the axis.*

Theorem 6.4 (Standing mass vs. travelling spin; `phasor3D_neutral_no_spin`, `phasor3D_charged_spin`, `resonantHeight_eq_exp_mellinReadoutFreq`)**.** *The neutral term is independent of the frequency y (`phasor3D_neutral_no_spin`): a* standing *(resonant) mode, not a travelling one, storing positive amplitude $n^{-\sigma}$ on the mass axis (`phasor3D_neutral_mass_pos`) while contributing* 0 *to the spin-plane L-series (`phasor3D_neutral_plane_zero`). The charged terms rotate at the readout rate* $\log n$ *as y advances (`phasor3D_charged_spin`) — the Mellin/readout frequency, distinct from the geometric carrier spin $n \cdot (\pi/3)$. The standing mass resonates at this readout frequency* $\mathrm{mellinReadoutFreq}(n) = \log n$ *and height* $\mathrm{resonantHeight}(n) = n = e^{\log n}$ *(`resonantHeight_eq_exp_mellinReadoutFreq`), matching the carrier height law $z_n = n = e^{y_n}$ of Remark 3.2.*

Remark 6.5 (Three channels, one decomposition; `phasor3D_three_channel_form`)**.** *Every term lies in exactly one channel — positive ($\chi(n) = +1$), negative ($\chi(n) = -1$), neutral ($\chi(n) = 0$) — and is determined by it (`phasor3D_three_channel_form`). The signed pair (positive/negative) is the spinning content the L-series sees; the neutral channel is the mass the planar model omits. This positive/negative/neutral split is the channel structure built into the harmonic pencil of Section 13.*

7 Strip extension: exact accumulation into $\mathrm{Re}\, s > 0$

The fiber does not merely approximate the analytic object near the critical line. After Abel summation it has an exact limit throughout the half-plane $\mathrm{Re}\, s > 0$: for non-principal characters that limit is $L(s, \chi)$ itself, and for the principal channel it is the exact eta-regularized zeta value. Thus the critical-line readout is reached from a proved strip identity, not from a finite truncation.

Theorem 7.1 (Dirichlet strip extension; `dirichlet_strip_tendsto_LFunction`). *For a Dirichlet character $\chi \neq 1$ modulo q and* $\operatorname{Re} s > 0$*,*

$$\sum_{n<N} \chi(n)\, n^{-s} \xrightarrow[N\to\infty]{} L(s,\chi).$$

The convergence target is exactly Mathlib's Dirichlet L-function, not an auxiliary regularization.

Theorem 7.2 (Eta strip extension; `eta_strip_tendsto`). *For* $\operatorname{Re} s > 0$ *and* $s \neq 1$*,*

$$\sum_{n<N} (-1)^{n+1}\, n^{-s} \xrightarrow[N\to\infty]{} \big(1 - 2^{1-s}\big)\zeta(s).$$

Proof sketch. Bucket cancellation gives a uniform bound on the character sums $\sum_{n<N}\chi(n)$; Abel summation by parts [6, 3] then yields convergence of $\sum \chi(n)n^{-s}$ on $\operatorname{Re} s > 0$, and the limit agrees with $L(s,\chi)$ on $\operatorname{Re} s > 1$ and hence everywhere on the connected strip by the identity theorem. For the alternating weight the factor $1 - 2^{1-s}$ is the even-term correction relating η and ζ. □

Theorem 7.3 (Continuous reach to the line; `continuous_model_zeta`). *For every* $y \in \mathbb{R}$*, at* $s = \frac{1}{2} + iy$*,*

$$\sum_{n<N} (-1)^{n+1}\, n^{-s} \longrightarrow \big(1 - 2^{1-s}\big)\zeta\big(\tfrac{1}{2} + iy\big),$$

and the limit is 0 *if and only if* $\zeta(\frac{1}{2} + iy) = 0$*.*

Proof sketch. Theorem 7.2 at $s = \frac{1}{2} + iy$ (using $\operatorname{Re} s = \frac{1}{2} > 0$ and $s \neq 1$). On the line the factor $1 - 2^{1-s}$ is nonzero (`correction_factor_ne_zero_on_critical_line`), so the product vanishes exactly when $\zeta(\frac{1}{2} + iy)$ does (`etaTrivial_eq_zero_iff`). □

7.1 Per-height convergence: the cancellation is in the fiber, not the carrier

The accumulation converges height-by-height, and its vanishing is a property of the *fiber*, never of the carrier. The two objects are distinct. The carrier $\mathrm{helixPt}(n) = \sqrt{n}\,\mathrm{wind}(n)$ has modulus exactly $\sqrt{n}$, independent of the ordinate y and the character χ (`carrier_norm_height_independent`), and never vanishes off the origin (`carrier_ne_zero`): it is a no-radial-drift source. The fiber paired against it is the unit-modulus spin (`fiber_spin_unimodular`) times the arithmetic datum $\chi(n)\, n^{-1/2}$; attaching it only *rotates* the carrier and cannot alter the $\sqrt{n}$ radial profile. Hence every vanishing of an accumulation is a fiber/pairing cancellation (orthogonality, Proposition 12.8), never a collapse of the carrier.

Theorem 7.4 (Per-height channels; `finiteCarrier_critical_tendsto`, `eta_critical_tendsto`, `perHeight_channel_nonprincipal`, `perHeight_channel_zeta`). *For every ordinate* $y \in \mathbb{R}$*:*

(i) *for non-principal* χ *the raw weighted partial sums* $\sum_{n<N}\chi(n)\, n^{-(1/2+iy)}$ *converge to* $L(\frac{1}{2} + iy, \chi)$ *(`finiteCarrier_critical_tendsto`), so the convergent channel's nonvanishing factor is* 1*;*

(ii) *for the principal character (and* ζ*) the raw series diverges, but the alternating (eta) channel* $\sum_{n<N}(-1)^{n+1}n^{-s}$ *converges to* $(1-2^{1-s})\zeta(s)$*, whose correction factor is nonzero on the line (`eta_critical_tendsto`, `zeta_eta_factor_ne_zero`).*

In both cases the channel value vanishes exactly when the L-value does (`LFunction_critical_eq_zero_iff_pairing`, `riemannZeta_critical_eq_zero_iff_eta_tendsto_zero`), the on-line limit $y \mapsto L(\frac{1}{2}+iy,\chi)$ is continuous (`continuous_LOnLine`), and it is the analytic limit of the finite carrier (`finiteCarrier_tendsto_LOnLine`).

This is the precise, defensible form of "every zero on the line is subject to the cancellation machine": it asserts nothing about *where* the zeros are, only that each on-line zero is exactly a fiber cancellation of the convergent channel.

7.2 A fully convergent worked fibre for ζ

For the Riemann zeta channel the convergence above is realized as an explicit, quantitatively controlled Cauchy sequence. Place the fibre at the critical-line point $\mathrm{sFiber}(T)$, which has real part $\frac{1}{2}$ by construction (`sFiber_re`).

Theorem 7.5 (The η-fibre converges on the line; `Pi3Fiber.PartialFiber_tendsto_FullFiber`, `PartialFiber_cauchySeq`, `FullFiber_eq_zero_iff_zeta`). *The alternating partial sums* $\mathrm{PartialFiber}(N,T) = \sum_{n<N}(-1)^n\,(n+1)^{-\mathrm{sFiber}(T)}$ *form a Cauchy sequence (`PartialFiber_cauchySeq`) converging to* $\mathrm{FullFiber}(T) = \eta(\mathrm{sFiber}(T))$ *(`PartialFiber_tendsto_FullFiber`, with vanishing tail `TailFiber_tendsto_zero`); and it vanishes* iff $\zeta(\mathrm{sFiber}(T)) = 0$ *(`FullFiber_eq_zero_iff_zeta`). The phasor form of the partial sum exhibits the spin* $\exp(-(HT\log(n+1))i)$ *on the amplitude* $(n+1)^{-1/2}$ *(`PartialFiber_eq_phasor`).*

Proof sketch. The summand differences are controlled by the consecutive-power bound $\left\|a^{-w}-(a+1)^{-w}\right\| \le \|w\|\,a^{-\operatorname{Re}w-1}$ for $a \ge 1$, $\operatorname{Re}w > 0$ (`cpow_neg_consecutive_norm_le`), giving summability of the bracketed series (`pairedTerm_summable`) and the Cauchy property on $\operatorname{Re}s > 0$. The bracketed sum agrees with η for $\operatorname{Re}s > 1$ and extends by the identity theorem across the preconnected region $\{\operatorname{Re}s > 0\}\setminus\{1\}$ (`pairedEta_eq_etaTrivial`, `rightHalfPlane_sdiff_one_isPreconnected`, `sFiber_ne_one`). □

Remark 7.6 (Exact accumulation, not approximate evaluation). *The critical-line behavior of L is classically reached by* approximate *means — the approximate functional equation and the Riemann–Siegel formula — which represent $L(\frac{1}{2}+it)$ by truncated sums with a controlled error term. The accumulation here is* exact*: the regularized carrier converges to L on the line (Theorems 7.4, 7.5) and vanishes* iff *L does (Theorem 9.5), an identity with no error term, machine-checked. Correspondingly the vanishing is an exact* event *— the sampled eigenwave is exactly ℓ^2-orthogonal to the arithmetic data (Proposition 12.8) — rather than an approximate minimum of $|L|$. The exactness is what lets the carrier's geometric and spectral structure transfer to L rigorously rather than heuristically; it is the foundation the rest of the development stands on. (The underlying convergence is the classical Abel/η summation on* $\operatorname{Re}s > 0$*; what is new is the exact, machine-checked geometric carrier built on it.)*

8 The geometric explicit formula

The logarithmic derivative of the fiber is the prime-power accumulation weighted by the winding.

Theorem 8.1 (Prime field; `explicit_formula_prime_field`, `dirichlet_logDeriv_eq_tsum`). *For* $\operatorname{Re}s > 1$,

$$-\frac{L'(s,\chi)}{L(s,\chi)} \;=\; \sum_n \chi(n)\,\Lambda(n)\,n^{-s},$$

where Λ is the von Mangoldt function. The zeros of $L(\cdot,\chi)$ are the poles of the meromorphic continuation of this object.

Definition 8.2 (Residue harmonic; `Residue.residueHarmonic`). *For $x > 0$ and $\gamma \in \mathbb{R}$, write $\rho = \frac{1}{2} + i\gamma$ and set* $\mathrm{residueHarmonic}(x, \gamma) = -x^\rho/\rho$.

Theorem 8.3 (Zeros located as poles; `zeta_zero_located_as_pole`, `L_zero_located_as_pole`). *Let $\rho = \frac{1}{2} + i\gamma$ be a zero of $L(\cdot,\chi)$ at which $L'(\rho,\chi) \neq 0$ (a simple zero). Then ρ is a simple pole of the explicit-formula kernel $-(L'/L)(s)\,x^s/s$, with*

$$\lim_{s\to\rho}(s-\rho)\left(-\tfrac{L'(s,\chi)}{L(s,\chi)}\cdot\tfrac{x^s}{s}\right) \;=\; \mathrm{residueHarmonic}(x,\gamma) \;=\; -\frac{x^\rho}{\rho}.$$

The same statement holds verbatim for ζ.

Proof sketch. On the punctured neighborhood of ρ in $\{s \neq 1\}$, L is analytic with $L(\rho) = 0$ and $L'(\rho) \neq 0$, so $-(L'/L)$ has a simple pole at ρ with residue 1; multiplying by the analytic factor x^s/s scales the residue to x^ρ/ρ and the leading sign gives $-x^\rho/\rho$. ☐

9 Projection and reality on the critical line

The carrier is three-dimensional; the readout projects it to a one-dimensional coordinate. We first record that coordinate and the change of scale it carries (Section 9.1), then the reality of the completed object that the readout returns (Section 9.2).

9.1 The readout coordinate and its scaling

The carrier is built in its native gauge: the unit is $\pi/3$, and a full angular turn is six such units, $6\cdot(\pi/3) = 2\pi$ (`spin_phasor_period6`, Theorem 3.13). The readout that sends a three-dimensional harmonic value $t \in \mathbb{R}$ down to a one-dimensional coordinate composes a Cayley map [9] with a rescaling to the standard unit interval (`HarmonicProjection`):

$$\mathrm{toCircleAngle}(t) = 2\arctan t \in (-\pi,\pi), \qquad \mathrm{toLine}(\theta) = \frac{\theta+\pi}{2\pi} \in (0,1),$$

and projection = toLine∘toCircleAngle. The second map carries the change of scale: it sends the full turn 2π to length 1, so the six $\pi/3$ buckets of the gauge become six equal sub-intervals of the readout. Under this rescaling the coordinate is rigid — the Cayley map is odd (`toCircleAngle_odd`), the normalization is affine hence midpoint-preserving (`toLine_midpoint`), and the principal-arc ends $\theta = \pm\pi$ map to the two ends of the interval (`toLine_neg_pi`, `toLine_pi`) — so the centre of the gauge is carried to the centre of the interval.

Theorem 9.1 (Coordinate centre; `HarmonicProjection.projection_midline`). projection(0) *is the midpoint of the readout interval: the centre of the six gauge buckets, the angle $\theta = 0$, maps to the average of the two endpoint images (`toLine_midpoint_of_endpoints`).*

The inverse conversion is the same scaling read backwards: a readout coordinate u is the angle $\theta = 2\pi u - \pi$, the gauge reading $6u - 3$ in $\pi/3$ units, and the interval's centre is the gauge centre $\theta = 0$. This is a property of the readout coordinate and its scaling. The value the readout returns on the completed object is real — it lands on the real axis (Section 9.2).

9.2 Reality on the line

The supporting analytic facts are the conjugation (Schwarz reflection [9]) and functional symmetries of Λ. The $\frac{1}{2}$ in this subsection is the *reflection axis* $\operatorname{Re} s = \frac{1}{2}$ — the fixed line of $s \mapsto 1 - s$ on which $\overline{s} = 1 - s$ (Section 2) — *not* the scale-critical exponent of Section 3.3: the reality is read off the functional equation, not the area law.

Theorem 9.2 (Conjugation symmetry; `HelixCollapse.completedRiemannZeta_conj`). *For all $s \in \mathbb{C}$, $\overline{\Lambda(\overline{s})} = \Lambda(s)$.*

Proof sketch. On $\operatorname{Re} s > 1$ this is the real Dirichlet coefficients of Λ ($\overline{\Lambda(\overline{s})} = \Lambda(s)$ termwise, using $\overline{\pi^{-s/2}} = \pi^{-\overline{s}/2}$ and $\overline{\Gamma(\overline{s}/2)} = \Gamma(s/2)$); the identity extends to $\mathbb{C}$ by analytic continuation of Λ_0 (the pole-removed completion) via the identity theorem. $\square$

Theorem 9.3 (Reality on the line; `completedRiemannZeta_critical_line_im_zero`). *For every $t \in \mathbb{R}$, $\operatorname{Im} \Lambda(\frac{1}{2} + it) = 0$.*

Proof sketch. On the line $\overline{\frac{1}{2} + it} = 1 - (\frac{1}{2} + it)$, so Theorem 9.2 together with $\Lambda(1 - s) = \Lambda(s)$ (`completedRiemannZeta_one_sub`) gives $\overline{\Lambda(\frac{1}{2} + it)} = \Lambda(\frac{1}{2} + it)$, i.e. a real value. $\square$

Theorem 9.4 (Faithful projection for ζ; `faithful_projection_zeta`). *For every $\gamma \in \mathbb{R}$,*

$$\operatorname{Im} \Lambda\big(\tfrac{1}{2} + i\gamma\big) = 0 \qquad \textit{and} \qquad F_\eta(\gamma) = 0 \iff \zeta\big(\tfrac{1}{2} + i\gamma\big) = 0,$$

where $F_\eta(\gamma) = (1 - 2^{1-s})\zeta(s)$ at $s = \frac{1}{2} + i\gamma$ (`EtaTrivial.Feta`).

Theorem 9.5 (Located crossings lie on the line; `crossing_is_zero_on_line`). *If the eta-regularized limit of Theorem 7.3 vanishes at ordinate y, then $\zeta(\frac{1}{2} + iy) = 0$ and $\operatorname{Re}(\frac{1}{2} + iy) = \frac{1}{2}$.*

10 Faithfulness for every Dirichlet L-function and Tate completion

A single carrier serves all χ; the character weight $\chi(n)$ on each phasor is the only dial.

Theorem 10.1 (One carrier, all characters; `faithful_all_L`). *For every Dirichlet character χ:*

(i) *for $\operatorname{Re} s > 1$, $\sum_{n<N} \chi(n)\, n^{-s} \to L(s, \chi)$; and*

(ii) *on the critical line $\operatorname{Re} s = \frac{1}{2}$, the eta-twisted closed carrier `etaTwistClosed`(χ, s) vanishes if and only if $L(s, \chi) = 0$ (`etaTwistClosed_eq_zero_iff_critical`).*

Definition 10.2 (Completed carrier; `Tate.completedCarrier`). *For a Dirichlet character χ, set $\Lambda^{c}_{\chi}(y) = \Lambda(\chi, \frac{1}{2} + iy)$ using the archimedean Γ-completion $\Lambda(\chi, s) = (\text{gamma factor}) \cdot L(s, \chi)$.*

Theorem 10.3 (Completion and functional equation; `faithful_all_L_completed`, `completedCarrier_eq_zero_i` `completed_functional_equation`). *For every* primitive *χ modulo q:*

(i) *the completion is zero-free off the zeros of L on the line: $\Lambda^{c}_{\chi}(y) = 0 \iff L(\frac{1}{2} + iy, \chi) = 0$ (the Γ-factor never vanishes for $\operatorname{Re} s > 0$); and*

(ii) *Tate's functional equation [12] holds: $\Lambda(\chi, 1 - s) = q^{\,s-1/2}\, W(\chi)\, \Lambda(\chi^{-1}, s)$, where $W(\chi) =$ `rootNumber`(χ).*

11 The double-ended carrier and conjugacy

The carrier is chiral: the right strand winds with spin $e^{-iy\log n}$, and its mirror, the left strand, winds with $e^{+iy\log n}$.

Theorem 11.1 (Conjugate chiralities; `both_helices_conjugate`)**.** *For every $N \in \mathbb{N}$ and $y \in \mathbb{R}$,*

$$\sum_{n<N} (-1)^{n+1}\, n^{-(1/2-iy)} \;=\; \overline{\sum_{n<N} (-1)^{n+1}\, n^{-(1/2+iy)}}\,.$$

Proof sketch. Termwise: $\overline{(-1)^{n+1}} = (-1)^{n+1}$ and $\overline{n^{-(1/2+iy)}} = n^{-(1/2-iy)}$ for the positive real base n (so $\arg n \neq \pi$), by `Complex.cpow_conj` and $\overline{n} = n$. □

Corollary 11.2 (Equal modulus, identical crossing heights)**.** *The two strands have equal modulus at every N, so their limits vanish at the same ordinates y.*

Theorem 11.3 (Completed sides agree; `both_sides_cross_together`)**.** *For every $y \in \mathbb{R}$, $\Lambda(\frac{1}{2} - iy) = \Lambda(\frac{1}{2} + iy)$.*

Proof sketch. $\frac{1}{2} - iy = 1 - (\frac{1}{2} + iy)$, so this is $\Lambda(1-s) = \Lambda(s)$ (`completedRiemannZeta_one_sub`) on the line. □

12 The Frobenius similitude and the determinant identity

The map $n \mapsto pn$ — and more generally $n \mapsto mn$ — is the *Frobenius self-similarity* of the carrier. We define it precisely, as a similitude of the carrier line.

Definition 12.1 (Frobenius similitude; `frobeniusMultiplier`, `frobeniusSimilitude`)**.** *For an integer m the* Frobenius multiplier *is $\mu_\theta(m) = \sqrt{m}\,\mathrm{wind}_\theta(m) \in \mathbb{C}$, and the* Frobenius similitude *$\mathrm{Sim}_\theta(m)$ is the $\mathbb{C}$-linear self-map $w \mapsto \mu_\theta(m)\, w$ of the carrier line $\mathbb{C}$.*

Theorem 12.2 (Prime multiplication acts by similitude; `helixPt_mul`, `helixPt_eq_similitude`, `norm_frobeniusMultiplier`, `frobeniusSimilitude_mul`)**.** *For $m, n \neq 0$, multiplication $n \mapsto mn$ acts on the carrier point $\mathrm{helixPt}_\theta(n) = \sqrt{n}\,\mathrm{wind}_\theta(n)$ by the similitude*

$$\mathrm{helixPt}_\theta(mn) \;=\; \mathrm{Sim}_\theta(m)\big(\mathrm{helixPt}_\theta(n)\big) \;=\; \underbrace{\sqrt{m}}_{radial}\;\underbrace{\mathrm{wind}_\theta(m)}_{rotation}\;\mathrm{helixPt}_\theta(n).$$

Its multiplier has modulus $\|\mu_\theta(m)\| = \sqrt{m}$ — the area-law factor of Theorem 3.9, the winding rotation being unit-modulus — so the map scales every length by $\sqrt{m}$. Moreover the similitudes compose*, $\mathrm{Sim}_\theta(mn) = \mathrm{Sim}_\theta(m) \circ \mathrm{Sim}_\theta(n)$: the carrier is* self-similar *under the whole multiplicative monoid generated by the primes (the Frobenius maps).*

Under the bridge (Theorem 4.4), with prime-angle assignment $\theta(p) = -y\log p$ the rotation eigenphase at ordinate y is the spin $z = \mathrm{spin}(y,n) = e^{-iy\log n} \in \mathbb{T}$ (`frobeniusMultiplier_bridge`). The two chiralities (Theorem 11.1) carry z and $\overline{z}$.

Theorem 12.3 (Conjugate-eigenstate determinant; `frobenius_conjugate_det_one`)**.** *For every $y \in \mathbb{R}$ and $n \in \mathbb{N}$, with $z = \mathrm{spin}(y,n) = e^{-iy\log n}$,*

$$\det\begin{pmatrix} z & 0 \\ 0 & \overline{z} \end{pmatrix} \;=\; z\,\overline{z} \;=\; |z|^2 \;=\; 1.$$

Proof. The 2×2 determinant is $z\cdot\overline{z}-0\cdot 0=z\,\overline{z}$. Since $\overline{z}=\overline{e^{-iy\log n}}=e^{\overline{-iy\log n}}=e^{+iy\log n}$ and $-iy\log n+\overline{-iy\log n}=0$, we have $z\,\overline{z}=e^{-iy\log n}\,e^{+iy\log n}=e^0=1$. □

Theorem 12.4 (Unitary transverse block; `frobeniusBlock_det_one`, `frobeniusBlock_unitary`). *The transverse block* $\mathrm{diag}(z,\overline{z})$ *assembled from the two chiral eigenphases at a conjugate crossing is* unitary, $M^{\mathsf{H}}M=I$, *with determinant* $z\,\overline{z}=|z|^2=1$ *— the orientation- and volume-preserving chiral invariant of the transverse phase dynamics. The radial scaling* $\sqrt{m}$ *of the similitude (Theorem 12.2) is carried separately and is not part of this block.*

12.1 The self-adjoint eigenstate

The determinant identity above is stated for the bare eigenphases $z,\overline{z}\in\mathbb{T}$. We realize them as the values of a genuine eigenstate of a self-adjoint generator. This is the operator *design*: it is real because its inputs are real, and it stands independently of any zero. Reading the nontrivial zeros as the spectrum of a self-adjoint operator is the *Hilbert–Pólya* program — supported by the random-matrix law of the zero statistics [13] and pursued through the Berry–Keating $H=xp$ quantization [14] and Connes's noncommutative-geometry trace formula [15]; the operator design here is unconditional and per-height, and makes no claim to be that canonical operator (Remark 12.11).

Definition 12.5 (Spectral wave; `spectralWave`). *For* $\gamma\in\mathbb{R}$, $\psi_\gamma(t)=e^{i\gamma t}$ $(t\in\mathbb{R})$.

Theorem 12.6 (Unit-modulus eigenstate of $D=-i\,d/dt$; `spectralWave_eigen`, `spectralWave_norm`). *For all* $t\in\mathbb{R}$, $-i\,\psi'_\gamma(t)=\gamma\,\psi_\gamma(t)$ *and* $\|\psi_\gamma(t)\|=1$. *Thus* ψ_γ *is a unit-modulus eigenstate of the generator* $D=-i\,d/dt$ *with the real eigenvalue* γ.

The reality is structural, not an accident of ψ_γ. Multiplication by the real height, $\mathrm{vonNeumannOp}(\gamma)=\gamma{\cdot}\mathrm{id}$ on the fiber $\mathbb{C}$, is symmetric with eigenvalue γ (`vonNeumannOp_isSymmetric`, `vonNeumannOp_hasEigenvalue`); on the finitely-supported sequence space the real-diagonal operator $\mathrm{diagOp}(d)$ is symmetric (`diagOp_symmetric`), with the basis vectors as explicit eigenvectors carrying the real eigenvalues d_i (`diagOp_eigenvector`), and *every* eigenvalue is real (`diagOp_spectrum_real`) by von Neumann reality — a symmetric operator on a complex inner-product space has real eigenvalues (`symmetric_eigenvalue_real`). These are collected in `unconditional_frobenius_eigenstate` and `frobenius_spectralWave_eigenstate`. The eigenwave is a one-parameter unitary group, $\psi_\gamma(s+t)=\psi_\gamma(s)\,\psi_\gamma(t)$ with $\psi_\gamma(0)=1$ (`spectralWave_add`, `spectralWave_zero`): the unitary flow generated by the real spectral parameter γ (Stone form).

Proposition 12.7 (The chiralities are eigenstate values; `spin_eq_spectralWave`, `conj_spin_eq_spectralWave`). *For every* $y\in\mathbb{R}$ *and* $n\in\mathbb{N}$, $\mathrm{spin}(y,n)=\psi_y(-\log n)$ *and* $\overline{\mathrm{spin}(y,n)}=\psi_y(\log n)$.

The useful spectral content is not the mere existence of the eigenwave ψ_y — every real y gives one — but its *orthogonality* to the arithmetic data at a vanishing.

Proposition 12.8 (The accumulation is an eigenstate–data inner product; `fiberCarrierFinite_eq_inner`, `fiberCarrierFinite_eq_zero_iff_orthogonal`). *Split the critical-line phasor* $\chi(n)\,n^{-(1/2+iy)}$ *into the arithmetic datum* $\chi(n)\,n^{-1/2}$ *(the finitely-supported vector* $\mathtt{dataVec}(\chi,N)$*) and the eigenwave sample* $\overline{\mathrm{spin}(y,n)}=\psi_y(\log n)$ *(the vector* $\mathtt{waveVec}(y,N)$*). With the* ℓ^2 *inner product* $\langle f,g\rangle=\sum_n\overline{f(n)}\,g(n)$ *on the finitely-supported sequence space, the finite accumulation is their pairing,*

$$\sum_{n<N}\chi(n)\,n^{-(1/2+iy)}\;=\;\big\langle\,\mathtt{waveVec}(y,N),\;\mathtt{dataVec}(\chi,N)\,\big\rangle,$$

so it vanishes iff *the sampled eigenwave is ℓ^2-orthogonal to the arithmetic data (`fiberCarrierFinite_eq_zero_iff_`*
A vanishing point is exactly a real-frequency orthogonality event, exhibited at each N (and the pairing is the partial phasor sum $G_{\chi,N}(\frac{1}{2}+iy)$, which accumulates to the carrier value $L(\frac{1}{2}+iy,\chi)$; `inner_eq_finiteCarrier`). The Gram form of any finite family of such vectors is positive-semidefinite (`carrier_gram_nonneg`), the positive-kernel background for the de Branges evaluation of Section 17.3.

Theorem 12.9 (Determinant on the eigenstate values; `frobenius_spectralWave_det_one`)**.** *For every $y \in \mathbb{R}$ and $n \in \mathbb{N}$,* $\det \operatorname{diag}\big(\psi_y(-\log n),\, \psi_y(\log n)\big) = 1.$

Theorem 12.10 (Frobenius unimodularity of the produced eigenstate; `frobenius_eigenstate_det_one`)**.** *Fix $\gamma \in \mathbb{R}$, $n \in \mathbb{N}$, and $\psi : \mathbb{R} \to \mathbb{C}$, and* assume *the vanishing $\Rightarrow$ eigenstate link $\psi = \psi_\gamma$. Then ψ is a unit-modulus eigenstate of $D = -i\, d/dt$ with eigenvalue γ, and* $\det \operatorname{diag}\big(\psi(-\log n),\, \psi(\log n)\big) = 1.$

Remark 12.11 (The open link)**.** *We exhibit the eigenstate ψ_γ for every real γ unconditionally (Theorem 12.6). What is* not *proved is that a carrier vanishing at height γ* produces *this eigenstate; that hypothesis ($\psi = \psi_\gamma$) is the one input of Theorem 12.10, left open. We do not derive that the vanishings are the spectrum of a canonical operator.*

13 The harmonic pencil and the focal eigenheight

The positive/negative channels of the 3D phasor (Section 6) assemble into a 2×2 determinantal pencil whose rank-drop locates the zeros — built in the geometric height $Z = e^y > 0$ rather than in $\log n$ (the *no-log-trap* design). The readout dictionary is $t = \log Z$, and the critical readout $\mathrm{criticalReadout}(Z) = \frac{1}{2} + i \log Z$ has real part $\frac{1}{2}$ for every real Z (`criticalReadout_re`).

13.1 The scalar channel closure

Definition 13.1 (Signed and unsigned harmonic channels; `HarmonicCell.Achan`, `Bchan`, `Focal.Pchan`, `Mchan`)**.** *With amplitude $a_n(t) = (\pi/3)\, n^{-1/2} \exp(-it\log n)$, the positive and negative scalar channels are $P(t) = \sum_{\chi(n)=+1} a_n(t)$ and $M(t) = \sum_{\chi(n)=-1} a_n(t)$. The* signed *harmonic mode is $B_\chi(Z) = (\pi/3)\, L(\frac{1}{2} + i\log Z, \chi)$ — the Abel limit of the channel difference $P - M$. The factor $\pi/3$ is the exact six-cell unit of the carrier (`Ucell`); it is nonzero (`Ucell_ne_zero`), so multiplying by it preserves the zero set exactly. The* unsigned *mode is $A_\chi(Z) = L(\frac{3}{2} + i\log Z, \chi)$, read at the absolute-convergence abscissa $\sigma = \frac{3}{2}$ where it never vanishes (`Achan_ne_zero`).*

Theorem 13.2 (Channel closure $\iff$ L-zero; `Focal.channel_diff_tendsto`, `channel_closure_iff_L_zero`)**.** *The scalar channel difference is the readout phasor partial sum, with Abel limit $(\pi/3)\,\Phi_\chi(Z) = (\pi/3)\, L(\frac{1}{2} + i\log Z, \chi)$ (`channel_diff_tendsto`). Hence the channel closes — the difference $P(\log Z) - M(\log Z)$ tends to 0 — iff $L(\frac{1}{2} + i\log Z, \chi) = 0$ (`channel_closure_iff_L_zero`). The $\pi/3$ scaling is what makes this a literal equality of cancellation events: the harmonic channel is a fixed nonzero multiple of the exact L-fiber, not a numerically tuned proxy.*

13.2 The determinantal pencil

Theorem 13.3 (Harmonic pencil determinant; `HarmonicCell.harmonicPencil_det`, `harmonic_pencil_det_zero`
For the diagonal harmonic pencil

$$H_\chi = \begin{pmatrix} A_\chi & B_\chi \\ \mu A_\chi & \lambda B_\chi \end{pmatrix}, \qquad \det H_\chi = (\lambda - \mu)\, A_\chi B_\chi.$$

Under admissibility ($A_\chi \neq 0$, $\lambda \neq \mu$) the rank-drop $\det H_\chi = 0$ *is exactly the scalar L-zero closure* $L(\frac{1}{2} + i\log Z, \chi) = 0$.

Theorem 13.4 (Gram rank-drop is the same event; `gram_rank_drop_iff_det_cell_zero`, `gramH_rank_drop_iff_L_zero`, `helix_eigenheight_real`)**.** *For any cell pencil L^{c} the Gram matrix $G^{\mathrm{c}} = (L^{\mathrm{c}})^{\mathsf{H}} L^{\mathrm{c}}$ is Hermitian positive semidefinite with the same kernel as L^{c}, and in the square case* $\det G^{\mathrm{c}} = |\det L^{\mathrm{c}}|^2$*; so the Gram rank-drop* $\det G^{\mathrm{c}}_{\chi,h} = 0$ *is again exactly the L-zero closure (`gramH_rank_drop_iff_L_zero`). The generalized eigenheights of such a positive pencil are real Rayleigh quotients $Z = \langle v, Av\rangle / \langle v, Bv\rangle$ (`helix_eigenheight_real`, `helix_eigenheight_rayleigh`), the no-log-trap height diagonal being $Z_n = n$, not* $\log n$.

13.3 The pencil as a family, and the finite crutch-free layer

The pencil is not a single matrix but a *family* of harmonic Gram interlaced operators $\{H_\chi(\mu, \lambda)\}$ over the admissible calibrations, and the rank-drop event is calibration-independent: all admissible members drop simultaneously, exactly when the signed channel closes (`gram_rank_drop_calibration_independent`; the event is governed solely by B_χ, independent of the unsigned mode — `projection_primacy`). Each member's Gram is Hermitian positive semidefinite (`gramH_posSemidef`) and is *never the zero matrix*: the rank drops by exactly one — a single cancellation, never a simultaneous double one (`specGram_apply00`, `specGram_ne_zero`; Section 14).

The detector itself is channel-agnostic — pure matrix algebra on a supplied pair, with no L-function in the statement: for any $A \neq 0$ and $\lambda \neq \mu$, $\det H(A, B; \mu, \lambda) = 0 \iff B = 0$, and likewise for the Gram (`harmonic_pencil_det_zero_iff_channel_zero`, `harmonicGram_rank_drop_iff_channel_zero`, `harmonicGram_rank_drop_calibration_independent`). Two instantiations are formalized. The *analytic* channels are $A_\chi = L(\frac{3}{2} + i\log Z, \chi)$ and $B_\chi = (\pi/3) L(\frac{1}{2} + i\log Z, \chi)$, with admissibility unconditional at the absolute-convergence abscissa (`Achan_ne_zero`). The *finite* channels are the live phasor-bank sums

$$A^{\mathrm{fin}}_\chi(a, N) = \sum_{n=1}^{N} \|\chi(n)\|\, a_n, \qquad B^{\mathrm{fin}}_\chi(a, \varphi, N) = \sum_{n=1}^{N} \chi(n)\, a_n\, e^{i\varphi_n},$$

with *arbitrary* magnitude window a and phase profile φ; here admissibility is discharged by *positivity* — $\chi(1) = 1$ and $a_1 > 0$ force $A^{\mathrm{fin}} > 0$ — with no analytic input (`finiteA_pos`, `finiteA_ne_zero`), and the family theorems specialize verbatim (`finite_gramH_posSemidef`, `finite_gramH_rank_drop_iff_channel_zero`, `finite_gram_rank_drop_calibration_independent`). A separate module re-houses the pencil test entirely on the finite side — L appears only in the final-verification bridge: the finite channels converge to the analytic ones (`AchanFin_tendsto`, `BchanFin_tendsto`), the finite signed channel closes in the growth limit iff $L(\frac{1}{2} + i\log Z, \chi) = 0$ (`BchanFin_closure_iff_L_zero`), and the asymptotic finite-pencil rank-drop is equivalent to the L-zero (`finite_pencil_rank_drop_iff_L_zero`).

13.4 Admissibility of the source-height readout

Theorem 13.5 (Real source-height $\Rightarrow$ critical line; `Admissible.admissible_iff_re_half`, `admissible_real_height_implies_critical_readout`, `nontrivial_zero_represented`)**.** *A complex point is the readout* criticalReadout(Z) *of a real source-height Z* iff *it lies on the critical line* $\mathrm{Re} = \frac{1}{2}$ *(`admissible_iff_re_half`); the backward direction realizes it at $Z = e^{\mathrm{Im}\,\rho}$. Equivalently an off-line ρ admits* no *real source-height (`offline_not_admissible`): the height solving the readout is real iff* $\mathrm{Re}\,\rho = \frac{1}{2}$ *(`height_im`, its imaginary part being* $-(\mathrm{Re}\,\rho - \frac{1}{2})$*). Consequently every* *nontrivial* critical-line *zero is represented by an admissible real source-height Gram eigen-event* $Z = e^t > 0$ *(`nontrivial_zero_represented`).*

Remark 13.6 (Projection primacy — the completeness question; `ProjectionPrimacy`, `projectionPrimacy_iff_exhausts`, `zero_source_admissibility_of_projectionPrimacy`). *The readout reaches* exactly *the critical line for real* Z *(`criticalReadout_re`), so the 3D carrier's projection cannot land off-axis and cannot host an off-line zero; Theorem 13.5 represents every zero* presented on *the line. The* projection-primacy principle *(`ProjectionPrimacy`) asserts that this projection is* complete *— that its real-height readout exhausts the nontrivial zeros — and is proved equivalent to that exhaustion (`projectionPrimacy_iff_exhausts`), hence to GRH for* χ*; conditional on it,* every *nontrivial zero (with no on-line hypothesis) is a represented admissible eigen-event (`zero_source_admissibility_of_projectionPrimacy`, `RH_of_everyZeroAdmissible`). The same principle is the representability bridge `EveryZeroAdmissible`; these two are the* formalized *faces of the one undecided question (Remark 15.7). The de Branges* IsHB Λ *condition is the same RH content in the classical de Branges form (Remark 17.2), which this work connects to but neither formalizes nor relies on — open work in the de Branges program, not a conditional of this paper. The open question itself is posed, never assumed: RH fails only if a factor outside the 3D model produces an off-line zero, which the carrier provably cannot. As an unconditional complement, the perturbation (Rouché-type) stability of the focal zeros is fully proved: at a simple critical-line zero* γ *where the L-channel dominates the bridge error there is a unique nearby focal zero, with the explicit location bound* $|T_d - \gamma| \le 2\varepsilon/a$ *(`real_simple_zero_location`, `real_simple_zero_exists`, `real_simple_zero_unique`).*

14 The self-adjoint realization and the GRH equivalence

The spectral layer realizes the cancellation events as kernel events of a fixed operator and then isolates, exactly, the RH-strength statement. The 3D spectral-fiber readout converges to the analytic L-function (`spectral_fiber_is_Lfunction`); a zero is a kernel event of the configured spectral operator, and conversely (`spectral_zero_has_kernel`, `spectral_kernel_iff_zero`, `cancellation_equivalence`). The generator A_χ is the real-diagonal operator with eigen-heights $\log(n+1)$ — strictly increasing because *only one phasor is growing continuously at a time* (`logGen_strictMono`); this is the formal counterpart of the partial-absorption law of Section 16.1.

Theorem 14.1 (No double cancellation; `no_double_cancellation`, `cancellation_modes_subsingleton`, `cancellation_energy_not_split`). *At a fixed spectral parameter s, at most one basis mode is a kernel vector of* $\mathrm{specOp}\, A_\chi\, s$*: two cancelling modes are the same mode. A cancellation event concentrates on a single site; the cancelled energy cannot be split between two distinct modes. Correspondingly the harmonic Gram matrix is never the zero matrix — its* $(0,0)$ *entry is* $1 + \|\mu\|^2 \ge 1$ *(`specGram_apply00`) — so the pencil family of Section 13.3 drops rank by exactly one per event.*

Theorem 14.2 (A realized cancellation is on the critical line; `nontrivial_Aχ_kernel_no_offline`). *If s lies in the open critical strip and some basis mode is a kernel vector of* $\mathrm{specOp}\, A_\chi\, s$*, then* $\mathrm{Re}\, s = \frac{1}{2}$*. The mechanism is von Neumann reality: the kernel condition forces the spectral height* $\mathrm{specHeight}(s) = -i(s - \frac{1}{2})$ *to equal a real eigen-height, hence to be real.*

Theorem 14.3 (GRH from a self-adjoint realization; `GRH_from_nontrivial_Aχ_realization`, `GRH_from_spectral_exhaustion`). *If every nontrivial cancellation of* $L(\cdot, \chi)$ *is realized as a kernel of the fixed self-adjoint generator* A_χ*, then every nontrivial zero lies on the critical line — GRH for* χ*. The implication is unconditional; the realization hypothesis is* *the isolated RH-strength input. The twin formulation through the sign-flip mechanism is equivalent: on the critical line the realization holds outright, and a real-generator cancellation forces* $\operatorname{Re} s = \frac{1}{2}$ *(`nontrivial_spectral_cancellation_to_real_generator_of_re_half`, `realGeneratorCancellation_re_half`).*

Remark 14.4 (The two-sided package, and where GRH sits)**.** *Theorem 13.5 realizes every* critical-line *zero as an admissible real-height eigen-event, unconditionally. Theorem 14.2 shows any* realized *nontrivial cancellation is on the critical line, unconditionally. The single unproved statement between them — that every nontrivial cancellation* is *realized — is the projection-primacy principle, proved equivalent to the real-height exhaustion of the zeros (`projectionPrimacy_iff_exhausts`) and hence to GRH for* χ *(Remark 15.7). This is a reframing of GRH as a completeness property of an explicit, elementary operator; it is not evidence for GRH, and nothing in this paper decides it.*

15 Admissibility: the carrier has no native off-axis support

The constructions above combine into a single intrinsic, on-axis statement about the carrier's *state space* — sharper than "a phasor plot that finds the zeros": it reframes the critical line as the carrier's admissible state space, the object whose completeness is projection primacy (Remark 15.7). A point of the vertical line $s = \sigma + iy$ rides the carrier as a stable, scale-normalized fiber only when its amplitude $n^{-\sigma}$ reciprocates the emergent area-law radius $r_n \sim \sqrt{n}$ to a finite positive scale. We call such a σ an *admissible (scale-balanced) state.*

Definition 15.1 (Scale-balanced admissible state; `ScaleBalanced`)**.** *For a carrier* (p, r) *and abscissa* $\sigma \in \mathbb{R}$*,* σ *is* scale-balanced *when the scale-balance product has a positive finite limit:* $\exists L > 0,\ n^{-\sigma} r_n \to L$*, where* $r_n = \text{carrierRadius}(p, r, n)$ *is the emergent area-law radius (Definition 3.8, Theorem 3.9).*

Theorem 15.2 (The admissible exponent is forced, not chosen; `scaleBalanced_iff`, `scaleBalanced_setOf_eq`)**.** *For every carrier* (p, r) *with* $r > 0$*,* σ *is scale-balanced* iff $\sigma = \frac{1}{2}$*; equivalently* $\{\sigma \in \mathbb{R} : \sigma \text{ scale-balanced}\} = \{\frac{1}{2}\}$*. The critical abscissa is the* unique *exponent compatible with the arclength area law (Theorem 3.11), and the conclusion is gauge-independent: the gauge* (p, r) *fixes only the value of the limit, never which* σ *is admissible.*

Corollary 15.3 (No native off-axis support; `not_scaleBalanced_of_ne`)**.** *For* $\sigma \neq \frac{1}{2}$ *the fiber is* not *a scale-balanced state: its amplitude* $n^{-\sigma}$ *is not scale-normalized to the carrier (it decays faster than the radius grows, or outruns it). The carrier supports no admissible off-axis state.*

Remark 15.4 (Answering the "designed around $\frac{1}{2}$" objection)**.** *The exclusion of off-axis states is not built in:* $\frac{1}{2}$ *is* derived *from the arclength area law (Theorems 3.9, 3.11) as the unique exponent at which* $n^{-\sigma}$ *reciprocates* $r_n \sim \sqrt{n}$*. It is the* $\frac{1}{2}$ *of* $\sqrt{n} = n^{1/2}$*, not a posited abscissa.*

The functional equation supplies the global counterpart: the reflection $s \mapsto 1 - s$, which pairs the two chiral fibers, fixes exactly the critical line.

Theorem 15.5 (The reflection fixes exactly the critical line; `reflection_fixes_iff`)**.** *For* $s \in \mathbb{C}$*,* $\operatorname{Re}(1 - s) = \operatorname{Re} s \iff \operatorname{Re} s = \frac{1}{2}$*. Off-axis,* $s \mapsto 1 - s$ *sends an abscissa imbalance* σ *to the opposite imbalance* $1 - \sigma$ *— a dual* pair*, not a self-conjugate state — so the double-ended carrier closes only on the fixed locus* $\operatorname{Re} s = \frac{1}{2}$*.*

Theorem 15.6 (The hierarchy, assembled; `no_native_offaxis_support`)**.** *For every carrier (p, r) with $r > 0$, height y, and site n, the following hold simultaneously:*

(1) *the area law $r_n^2/n \to r(\pi/3)/\pi$ (so $r_n \sim \sqrt{n}$), from the arclength geometry;*

(2) *scale balance holds iff $\sigma = \frac{1}{2}$ (forced, not chosen);*

(3) *every $\sigma \neq \frac{1}{2}$ is inadmissible;*

(4) *at the balanced axis the chiral fibers $z, \overline{z}$ form a determinant-one* unitary *transverse block (Theorem 12.4); and*

(5) *$s \mapsto 1 - s$ fixes exactly $\operatorname{Re} s = \frac{1}{2}$, on which the de Branges variable is real (Corollary 17.1).*

The chain

$$r_n \sim \sqrt{n} \implies \sigma = \tfrac{1}{2} \implies \textit{unitary chiral block} \implies \det = 1 \implies s \leftrightarrow 1 - s \textit{ closes on } \operatorname{Re} s = \tfrac{1}{2}$$

is the carrier's intrinsic logic: it reads as a geometry, not a graphing of $L(\frac{1}{2} + it)$.

Remark 15.7 (Projection primacy: the pivotal hypothesis)**.** *Theorem 15.6 establishes that the carrier's admissible state space — the image of its real-height projection $Z \mapsto \frac{1}{2} + i\tau_\chi(\log Z)$ — is the critical line: off-axis points are unsupported / inadmissible / not scale-normalized, so the 3D representation* cannot *host an off-line zero (`criticalReadout_re`). Whether the projection is also* complete *— whether it exhausts* every *nontrivial zero — is the* projection-primacy principle *(`ProjectionPrimacy`), proved equivalent to the real-height exhaustion of the zeros (`projectionPrimacy_iff_exhausts`) and hence to GRH for χ; conditional on it the exhaustion is unconditional (`zero_source_admissibility_of_projectionPrimacy`). This is the open consideration, and it is a genuine dichotomy:* either *the projection is primary — the more accurate 3D model defines the 1D/2D readout completely, and there are no off-line zeros —* or *some factor outside the 3D representation produces an off-line zero, one the carrier provably cannot. The Riemann Hypothesis is thus not something weaker that this work falls short of, but exactly the completeness of the projection; the work poses the dichotomy and proves the off-axis branch impossible* within *the model. "Frobenius" here means precisely the prime-multiplication similitude of Definition 12.1.*

16 Computational phenomenology

The preceding sections are theorems. This section is science: the model run as an instrument, with its measurements, its error budget, and its negative results. All quantities below are reproduced by the repository harnesses (`phasor_explorer/model.py`, `focal_closure.py`, `carrier_fiber/carrier_fiber.py`) and the interactive WebGL explorer (`phasor_explorer/explorer_v2.html`), which renders up to 10^5 phasors with shader-evaluated phases and exposes every mode discussed here.

16.1 The partial-absorption law

The fiber starts empty at the carrier origin (the $n = 0$ site carries nothing; the bank lives at heights > 0), and each phasor is absorbed *continuously*: with head height H, phasor n carries the window

$$a_n(H) \;=\; n^{-1/2}\,\operatorname{clip}\bigl(H - n + 1,\, 0,\, 1\bigr),$$

so it starts growing as the head leaves site $n-1$ and completes exactly *at* its own integer site — one phasor partial at any instant, nothing discontinuous. At integer heads the absorbed bank equals the completed snapshot exactly, and between sites the signed sum moves along a straight segment whose velocity is the single absorbing phasor's term (α-linearity, verified to 10^{-12}). One-at-a-time absorption is load-bearing: it keeps the eigen-heights strictly monotone, which is the hypothesis chain of Theorem 14.1. A τ-window variant $a_n(H) = n^{-1/2}\mathrm{clip}((H-n)/\tau, 0, 1)$ — many phasors growing simultaneously — remains available as a smoothing device; it sharpens the focal residual (below) at the cost of the one-at-a-time structure.

16.2 Validation: all critical-line zeros below height 36, eight characters

For the eight real primitive characters of conductor $q \leq 12$ (the η-regularized principal channel and the quadratic characters of conductor $3, 4, 5, 7, 8, 11, 12$) we computed *every* critical-line zero with $\gamma < 36$ — 111 ordinates in total ($5, 11, 12, 14, 16, 16, 18, 19$ per conductor) — by Newton refinement on $L(s,\chi) = q^{-s}\sum_a \chi(a)\zeta(s, a/q)$ (Hurwitz-zeta evaluation, 22 significant digits, stored to 10 decimals). The bound $\gamma < 36$ is an integer-exactness cap, not a cost cap: the eigenheight $N = \mathrm{round}(e^{\gamma})$ must stay below 2^{53} for exact modular arithmetic.

At every one of the 111 pairs (γ_k, $N = e^{\gamma_k}$) the focal event of Theorem 13.2 was evaluated. With the absorption law (equivalently the completed snapshot, since N is an integer) the residual $|B|$ lies below the display threshold 0.05 at 108/111 zeros; the three exceptions are precisely the smallest banks — conductor 7 (γ_1, $N = 88$, $|B| = 0.107 \approx 1/\sqrt{88}$), conductor 8 ($N = 134$, 0.087), conductor 12 ($N = 45$, 0.153) — honest truncation at that scale, and the τ-window closes all three (0.008–0.016 at $\tau = 400$). At the deep end the residuals sit at the truncation scale $\sim N^{-1/2}$: for conductor 11, $\gamma_{18} = 35.4850851876$ gives $N = 2{,}576{,}182{,}862{,}426{,}122 \approx 2.6 \times 10^{15}$ and $|B| = 7.3 \times 10^{-11}$; for ζ, γ_4 gives $N \approx 1.6 \times 10^{13}$ and $|B| = 1.2 \times 10^{-7} \approx N^{-1/2}/2$.

16.3 The error budget

The residual at a focal event decomposes into three measurable floors. *Zero precision*: with ordinates stored to 6 decimals the deep residuals are dominated by the rounding of γ itself (conductor 11, γ_{18}: 6.2×10^{-7} at 6 decimals versus 7.3×10^{-11} at 10), so headline depths quote the data precision, not the model. *Truncation*: with the zeros at 10 decimals the residual tracks $\sim N^{-1/2}$ across ten orders of magnitude of N. *Evaluator*: beneath both lies the analytic evaluator's own floor ($\approx 10^{-9}$–10^{-10}, next subsection). Misattributing the first floor to the second is the kind of error this budget exists to prevent.

16.4 The $O(1)$ evaluator

Brute evaluation is $O(N)$ and the eigenheights reach 10^{15}. Every phase mode except the continuous carrier-rate comparison has residue-class structure, so the bank sum splits as an exact 6000-term head plus per-class Euler–Maclaurin tails (integral, boundary, and B_2, B_4, B_6 corrections, all closed-form for n^{-s}), with the absorption window contributing one explicit partial term. The cost is independent of N; the absolute error is $\sim 10^{-9}$. The evaluator is validated against the brute sum on 240 configurations (eight characters $\times$ three phase modes $\times$ five magnitude laws $\times$ two sizes, agreement $< 10^{-8}$) plus a 6.6×10^{6}-term brute anchor, and the JavaScript mirror agrees with the Python reference to nine decimals at fractional heads. This is what makes Section 16.2 a table instead of a supercomputer allocation.

16.5 Degree two: the mechanism does not require periodicity

The cheapest dismissal of Section 16.2 would be that character periodicity — complete residue blocks cancelling — secretly does all the work. It does not. The locator of `focal_closure.py` runs the same construction on degree-two L-functions, with the lanes given by *coefficient signs* instead of residue classes: for the Ramanujan form, $\lambda(n) = \tau(n)/n^{11/2}$ built exactly from $\eta(q)^{24}$ (with Hecke multiplicativity and Deligne-bound assertions checked at build time), and for the level-11 elliptic newform, $\lambda(n) = a_n/\sqrt{n}$ from $\eta(q)^2\eta(q^{11})^2$. The growth locator finds the first zero of $L(\Delta, s)$ unaided to 10^{-6} of the reference ordinate $9.2223793999\ldots$, with genuine closures reaching depth 10^{-6}–10^{-12} as the bank deepens; the level-11 zeros likewise. Neither form is periodic, neither has unimodular coefficients. Within the arrows the $\pi/3$ scale is provably inert (a constant amplitude and a global phase, cancelling between lanes — the gauge test of `focal_closure.py`); its load-bearing roles are exactly the two proved ones: the carrier placement forcing the area law (Theorem 3.9) and the exact cell normalization $B = (\pi/3)L$ (Theorem 13.2).

16.6 Negative results

Three natural conjectures about the carrier were tested and are null, and one phenomenological law is positive. (i) A $\pi/3 \cdot \mu_6$ carrier-pattern search for *close pairs* of zeros shows no enrichment: lift ≈ 1.0, $p \approx 0.9$. (ii) A pair-precursor signal (structure in the fiber preceding a close pair) is absent. (iii) A 6- or 12-fold "harmonic clock" in the zero sequence is absent; only the $k = 1$ harmonic is real. (iv) Positive: after each cancellation the fiber's imbalance *reopens* at a fixed linear rate ($p \approx 1.05$, uncorrelated with the gap to the next zero), with close (Lehmer-type) pairs entering through a quadratic crossover rather than any repulsion mechanism; the reverb kernels are formalized in `ReverbResidue.lean`. We report (i)–(iii) with the same confidence as the positive results: the carrier structure is a home for the cancellation event, not a hidden periodicity in the zeros.

16.7 The finite instrument

The finite Python harness `carrier_fiber/carrier_fiber.py` separates the carrier geometry from the analytic calibration. Its native finite path uses the pitch-one $\pi/3$ carrier directly: a height $z > 0$ absorbs

$$N = \lfloor z \rfloor$$

complete integer slots and the fractional part $\alpha = z - N$ of the next slot. With a real periodic coefficient $c(n)$, the signed finite fiber term is

$$v_n = c(n)\,((\pi/3)n)^{-1/2} \exp\bigl(i(n-1)\pi/3\bigr).$$

The whole-fiber focal value at height z is therefore the piecewise-linear path

$$B(z) = \sum_{n=1}^{N} v_n + \alpha v_{N+1}.$$

The readout ordinate printed by the harness is $\log z$; the height itself is the carrier coordinate.

The native scan `find_continuous_focal_crossings` finds candidate vanishings without a one-dimensional nonlinear root solver. On each interval $[N, N+1]$ it writes $S_N = \sum_{n=1}^{N} v_n$ and minimizes the distance from the origin to the segment $S_N + \alpha v_{N+1}$ by the explicit projection

$$\alpha_N = \mathrm{clip}_{[0,1]}\left(-\frac{\mathrm{Re}(S_N\overline{v_{N+1}})}{|v_{N+1}|^2}\right).$$

The residual is

$$r_N = \big|S_N + \alpha_N v_{N+1}\big|.$$

Heights with r_N below the requested tolerance are reported as exact finite focal crossings; if no exact crossing is present in the scan range, the harness reports the smallest residuals. In compact form:

```
terms = c(n) * ((pi/3) * n)**(-0.5) * exp(1j * slot * pi/3)
prefix[N] = sum_{n <= N} terms[n]
alpha[N] = clip(-real(prefix[N] * conj(terms[N+1])) / abs(terms[N+1])**2, 0, 1)
height[N] = N + alpha[N]
residual[N] = abs(prefix[N] + alpha[N] * terms[N+1])
```

The same scan also evaluates the harmonic pencil marker. The unsigned admissibility channel is

$$A(z) = \sum_{n=1}^{N} |c(n)|\,((\pi/3)n)^{-1/2} + \alpha|c(N+1)|\,((\pi/3)(N+1))^{-1/2}.$$

For each pencil

$$H(z) = \begin{pmatrix} A(z) & B(z) \\ \mu A(z) & \lambda B(z) \end{pmatrix},$$

the code checks

$$\det H = (\lambda - \mu)A(z)B(z), \qquad \det(H^*H) = |\det H|^2.$$

Thus, whenever $A(z) \neq 0$ and $\lambda \neq \mu$, the Gram rank-drop marker is exactly the same event as the finite focal cancellation $B(z) = 0$.

The default validation command uses the cumulative c1 carrier clock, which is the Lean-aligned readout used for the L-channel:

$$T = \gamma/(\pi/3), \qquad B_N(T) = \frac{\pi}{3} \sum_{n \le N} c(n) n^{-1/2} \exp\big(-i(\pi/3)T \log n\big).$$

In finite mode the harness evaluates this finite sum and locally minimizes $|B_N(T)|^2$ by a coarse grid followed by golden-section refinement in the window around $T = \gamma/(\pi/3)$. In analytic mode it uses the exact calibrated channels

$$A(T) = L(3/2 + i\gamma, \chi), \qquad B(T) = (\pi/3)L(1/2 + i\gamma, \chi),$$

then applies the same determinant and Gram determinant checks. The reference γ table selects which displayed ordinates to validate; the focal marker remains $B = 0$ and the pencil marker remains $\det H = 0$.

17 Structural frameworks (outlook)

Three machine-checked frameworks sit above the main line. They are scaffolding — correct, kernel-checked, and deliberately demoted here until they carry mathematical weight of their own.

17.1 The chiral cup

The forward/reverse chiralities of the double-ended carrier assemble a Hermitian, positive *definite* sesquilinear cup on the finitely-supported fibre space, which completes to a genuine Hilbert space with the ℓ^2 energy on the diagonal (`cup_hermitian`, `cup_null_iff`, `hilbert_completion_exists`); the Frobenius transport acts as a cup similitude scaling by $\|\mu\|^2$, unitary after normalization (`transport_unitary_after_normalization`), a crossing-induction argument propagates chiral modulus balance (`no_dominance_by_crossing_induction`), and on the prime-power probe the cup energy carries the von Mangoldt field, $-L'/L$ for $\operatorname{Re} s > 1$ (`cup_intertwining`, `vonMangoldtChannel_eq_neg_logDeriv`).

17.2 The Kähler polarization and the trace

The fibre carries a compatible triple (Ω, J, g) with the Kähler identity $\langle x, y\rangle = g(x, y) + i\,\Omega(x, y)$, strictly positive on the chiral defect (`inner_eq_gH_add_I_OmegaH`, `theoremA`); Frobenius transport scales the symplectic pairing by exactly p^r (`theoremB`), and the weighted local trace of the site-shift dynamics is the von Mangoldt superposition, equal to $-L'/L$ for $\sigma > 1$ (`theoremC`). A cohomological re-expression of the vanishing exists ($H^\bullet(\rho) = \mathbb{C}/\langle L(\rho, \chi)\rangle$; `theoremD`) but is, as formalized, the field-theoretic identity "$\mathbb{C}/\langle x\rangle \neq 0 \iff x = 0$" — we record it as notation awaiting content, not as a cohomology theory.

17.3 The de Branges evaluation

The Hermite–Biehler/de Branges framework is formalized generically: positivity of the structure kernel forces the HB inequality and hence a real, discrete spectrum (`featureMap_forces_HB`, `hb_no_zero_upper`, `Bcomp_zeros_discrete`), with the Paley–Wiener function e^{-iz} as the fully unconditional worked template (`paleyWiener_isHB`; spectrum $\{k\pi\}$).

Corollary 17.1 (On-line cancellations are real spectral points; `criticalLine_deBranges_real`, `zeroHarmonic_deBranges_real`)**.** *Under the change of variables $z = -i(\rho - \frac{1}{2})$, a carrier vanishing at height γ sits at the* real *de Branges point $z = \gamma$; reality of z is exactly* $\operatorname{Re}\rho = \frac{1}{2}$ *(`deBranges_var_im`).*

Remark 17.2 (Scope: $\mathrm{IsHB}\,\Lambda$ is classical open work, not used here)**.** *That a Λ-built structure function is Hermite–Biehler is equivalent to the Riemann Hypothesis — the classical de Branges reformulation. No such structure function is constructed here; only the unconditional on-line content (Corollary 17.1) is used. The open question of this paper is projection primacy (Remark 15.7),* not $\mathrm{IsHB}\,\Lambda$.

18 Formalization

Every definition and theorem above is formalized in Lean 4 [19] against Mathlib [18], in the package `RequestProject` (files `ClosedForm.lean`, `HelixLogFreeFTA.lean`, `HelixCollapseReality.lean`, `LFunctionPhasor.lean`, `GeometricProjectionHolds.lean`, `Faithfulness.lean`, `AreaLaw.lean`, `UnconditionalFrobenius.lean`, `DeBranges.lean`, `FrobeniusSimilitude.lean`, `PerHeightConvergence.lean`, `HeightGrowthActive.lean`, `CircleMonodromy.lean`, `CriticalLineBridge.lean`, `GeometricPhasorClosure.lea` `Phasor3D.lean`, `FocalEigenheight.lean`, `HarmonicPencilCell.lean`, `FinitePencil.lean`, `AdmissibleEigenheight.lean`, `FullFiber.lean`, `PrimePowerProbe.lean`, `CupIdentity.lean`, `ChiralCup.lean`, `CrossingInduction.lean`, `CrossingEncodingAudit.lean`, `HelixPolarization.lean`, ConditionalToUnconditional.lean, SpectralFiberIsLFunction.lean, SpectralZeroKernel.lean, CancellationEquivalence.lean, SelfAdjointGeneratorReadout.lean, SpectralSignFlip.lean, NoDoubleCancellation.lean, SpectralExhaustion.lean, SpectralGramEvent.lean, SpectralRealizationGR FocalResidualVanishes.lean, FocalCancellationFindsZeros.lean, FocalCancellationEigenstate.lean, GeometricReadout.lean, ReverbResidue.lean, and MeromorphicTraceIdentity.lean). Each named theorem compiles with no sorry; the kernel axiom footprint reported by #print axioms (equivalently lean_verify) is

$$\{\ \text{propext},\quad \text{Classical.choice},\quad \text{Quot.sound}\ \},$$

the three standard Mathlib axioms, with no construction-specific axiom. The correspondence between the statements here and the Lean declarations is given by the identifiers in parentheses throughout.

The capstone identities — the strip extension (Theorems 7.1, 7.2, 7.3), the conjugacy of the two chiralities (Theorem 11.1), and the determinant identity (Theorem 12.3) — are collected in Faithfulness.lean; the geometry (Section 3) and the log-free winding (Section 4) live in ClosedForm.lean and HelixLogFreeFTA.lean, with the emergent area-law radius (Theorem 3.9, Definition 3.8) and the scale-critical exponent (Theorem 3.11) in AreaLaw.lean. The self-adjoint eigenstate design (Section 12.1) lives in UnconditionalFrobenius.lean and the Hermite–Biehler / de Branges framework (Section 17.3) in DeBranges.lean, with their connection to the carrier — the eigenstate identification (spin_eq_spectralWave, frobenius_eigenstate_det_one) and the on-line de Branges evaluation (criticalLine_deBranges_real, zeroHarmonic_deBranges_real) — collected in Faithfulness.lean. The compact local spectral mechanism of Section 12 — the Frobenius similitude (Definition 12.1, Theorem 12.2), the unitary transverse block (Theorem 12.4), the eigenstate–data orthogonality (Proposition 12.8), and the admissibility / no-off-axis-support results of Section 15 (Theorems 15.2, 15.5, 15.6) — lives in FrobeniusSimilitude.lean, assembled in frobenius_local_spectral_mechanism and no_native_offaxis_support. The remaining layers map as follows: the per-height convergence on the line (Section 7.1) is PerHeightConvergence.lean; the constant height law and active fraction (Remark 3.2) and the exact $\pi/3$ monodromy (Theorem 3.15) are HeightGrowthActive.lean and CircleMonodromy.lean; the geometric phasor-closure dictionary and lane split (Section 7) are GeometricPhasorClosure.lean and CriticalLineBridge.lean; the three-channel 3D phasor (Section 6) is Phasor3D.lean; and the harmonic pencil, focal eigenheight, and source-height admissibility (Section 13) are HarmonicPencilCell.lean, FocalEigenheight.lean, and AdmissibleEigenheight.lean. The chiral cup and its Hilbert completion (Section 17.1) live in CupIdentity.lean, ChiralCup.lean, CrossingInduction.lean (with the non-vacuity audit CrossingEncodingAudit.lean) and PrimePowerProbe.lean; the Kähler polarization, the trace formula, and the cohomological re-expression of the zeros (Section 17.2) are HelixPolarization.lean; the convergent η-fibre (Section 7.2) is FullFiber.lean; and the unconditional Paley–Wiener template (Section 17.3) is ConditionalToUnconditional.lean.

The updated spectral/focal layer is split into smaller audit modules. SpectralFiberIsLFunction.lean proves that the 3D spectral-fiber readout converges to the analytic L-function (spectral_fiber_is_Lfunction) and that the eta-configured principal readout recovers ζ on the punctured strip (etaSpectralFiber_readout_recov SpectralZeroKernel.lean and CancellationEquivalence.lean package zeros as kernel events of the configured spectral operator (spectral_zero_has_kernel, spectral_kernel_iff_zero, cancellation_equivalence). The self-adjoint readout and its Gram markers live in SelfAdjointGeneratorReadout.lean, SpectralSignFlip.lean, NoDoubleCancellation.lean, SpectralExhaustion.lean, and SpectralGramEvent.lean: these record the spectral height `specHeight`, the signed mode `specBchan`, the rank-drop criterion `specGram_det_zero_iff`, sign-change and real-axis facts (`spectral_cancellation_sign_flip`, `sign_flip_only_on_real_axis`), distinct-mode exclusion (`no_double_cancellation`), and the critical-line implication from an L-zero to a harmonic Gram event (`L_zero_imp_spectral_gram_event`). The focal-cancellation modules `FocalResidualVanishes.lean`, `FocalCancellationFindsZeros.lean`, and `FocalCancellationEigenstate.lean` formalize that the normalized focal residual vanishes exactly with the scalar L-readout (`focal_residual_zero_iff_L_zero`), that an on-line zero at ordinate y is found at source height e^y (`focal_cancellation_finds_online_zero`), and that an exact charged focal cancellation produces a standing/eigenstate package with conserved magnitude (`focal_cancellation_produces_eigenstate`). `GeometricReadout.lean` collects the height-readout and no-double-residue facts, and `MeromorphicTraceIdentity.lean` records the completed trace identities for the spectral operator.

Reproduction. The Lean claims: `lake build` (Mathlib pinned by the manifest); axiom footprints via `#print axioms`. The phenomenology of Section 16: `python3 model.py` (absorption law, evaluator-vs-brute on 240 configurations, pencil family), `python3 focal_closure.py test` (growth locator; the degree-two families with Hecke/Deligne build-time checks), and `python3 carrier_fiber.py test` (finite carrier-clock harness). The interactive explorer (`phasor_explorer/explorer_v2.html`) exposes every character, every listed zero, and every magnitude law of this paper. Appendix A is the claim-by-claim map.

Remark 18.1 (Scope)**.** *This paper presents a geometric state space: the double-ended Frobenius helix carrier and its phasor fiber, whose admissible scale-balanced states exist only on the critical line (Section 15) and on which the vanishing points are* exactly *the on-line zeros of ζ and every Dirichlet L-function, by the formalized equivalence (Theorem 10.1, Section 7.2). Its central contribution is intrinsic: the carrier has* no native off-axis support *(Theorem 15.6), and RH is reframed as the* projection-primacy *principle — the completeness of the carrier's real-height projection (`projectionPrimacy_iff_exhausts`), equivalent to GRH, with the off-axis branch proved impossible inside the model. Its mathematical content is exactly the statements proved above — the carrier, the area-law derivation of the critical exponent (Theorem 3.11), the accumulation to $L(s,\chi)$ on* $\operatorname{Re} s > 0$*, the explicit-formula reading of the zeros as poles, the reality of the completed object on the critical line, the conjugacy of the two chiralities, the Frobenius similitude (Theorem 12.2), the unitary transverse block (Theorem 12.4), the eigenstate–data orthogonality (Proposition 12.8), the self-adjoint eigenstate realization of the chiral eigenphases (Section 12.1), the de Branges evaluation placing the on-line cancellations as real spectral points (Section 17.3), and the admissibility restriction of the carrier's state space to the critical line (Section 15). The pivotal hypothesis is* projection primacy *— the completeness of the 3D-to-1D projection (Remark 15.7) — formalized in two guises: that the real-height readout exhausts the zeros (`projectionPrimacy_iff_exhausts`), and that a vanishing* produces *the eigenstate (Remark 12.11). The classical de Branges* IsHB Λ *condition (Remark 17.2) is the same RH content in the de Branges form — connected here but neither formalized nor relied on (open work, not a conditional of this paper). This work does not decide projection primacy: it poses the dichotomy — the projection is primary, so there are no off-line zeros,* or *an unmodeled factor produces one — and proves that the off-axis branch cannot arise within the 3D carrier. The Riemann Hypothesis is exactly the resolution of this dichotomy, which this work frames but neither assumes nor proves.*

A Headline claims and their Lean identifiers

Each row is a claim of this paper, the Lean identifier that proves it, and its file (package `RequestProject`, 43 modules). All rows compile with no `sorry`; axiom footprint $\{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$.

Claim	**Lean identifier**	**File**
Area law $r_n^2/n \to r\Delta/\pi$	`windIntegerSite_radius_sq_tendsto`	`AreaLaw`
$\sigma = \frac{1}{2}$ uniquely scale-balanced	`sigma_half_is_scale_critical`, `scaleBalanced_iff`	`AreaLaw`
$\pi/3$ has exact finite monodromy	`pi3_finite_monodromy`	`CircleMonodromy`
Strip convergence to $L(s,\chi)$, $\operatorname{Re} s > 0$	`dirichlet_strip_tendsto_LFunction`	`LFunctionPhasor`
η-channel identifies ζ's on-line zeros	`FullFiber_eq_zero_iff_zeta`	`FullFiber`
Vanishing = eigenwave–data orthogonality	`fiberCarrierFinite_eq_zero_iff_orthogonal`	`FrobeniusSimilitude`
Channel closure $\iff$ L-zero	`channel_closure_iff_L_zero`	`FocalEigenheight`
Pencil det $= (\lambda - \mu)AB$; rank-drop $\iff$ L-zero	`harmonicPencil_det`, `gramH_rank_drop_iff_L_zero`	`HarmonicPencilCell`
Rank-drop is calibration-independent	`gram_rank_drop_calibration_independent`	`HarmonicPencilCell`
Channel-agnostic detector (no L in statement)	`harmonicGram_rank_drop_iff_channel_zero`	`HarmonicPencilCell`
Finite admissibility by positivity	`finiteA_pos`, `finite_gramH_rank_drop_iff_channel_zero`	`HarmonicPencilCell`
Finite pencil $\to$ L only at final verification	`finite_pencil_rank_drop_iff_L_zero`	`FinitePencil`
Every critical-line zero realized at $Z = e^{\gamma}$	`nontrivial_zero_represented`	`HarmonicPencilCell`
One cancellation site per event	`no_double_cancellation`	`NoDoubleCancellation`
Realized cancellation $\Rightarrow$ on the line	`nontrivial_A`χ`_kernel_no_offline`	`SpectralRealizationGRH`
Realization of all cancellations $\Rightarrow$ GRH	`GRH_from_nontrivial_A`χ`_realization`	`SpectralRealizationGRH`
Projection primacy $\iff$ real-height exhaustion (= GRH)	`projectionPrimacy_iff_exhausts`	`HarmonicPencilCell`
No native off-axis support	`no_native_offaxis_support`	`FrobeniusSimilitude`
Reality of Λ on the line	`completedRiemannZeta_critical_line_im_zero`	`HelixCollapseReality`
Reverb reopening kernels	`ReverbResidue` (module)	`ReverbResidue`

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